Sensors for Obstacle Detection – A Survey

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Abstract: *The obstacle detection field is a very broad one and a lot of obstacle detection systems have been developed in the last years in this domain. We tried to identify the main character of an obstacle detection system from the ruttier scene. Thus, we classified the main types of sensors from this field in passive (visible and infrared spectrum camera) and active (radar, laser-scanner, sonar) sensors and we made a survey in this domain. After a short presentation of every type of sensor, we presented another current and fancy solution for an obstacle detection system: the fusion of different sensor together. Almost all obstacle detection systems use a combination of passive-active technology, and in general the best solution is obtained using a vision system combined with a distance sensor like radar or laser. Maybe the most low-priced system is one combining only vision systems, but the inconvenient in this case is the lack of distance information.*

1. Introduction

Objects to be detected in a ruttier scene have a large variety of appearances due to different shapes, sizes, orientation with respect to camera's position. Combined with a cluttered background and a range of visibilities under various lighting and weather conditions, the solution for a reliable and accurate algorithm for identifying an obstacle requires complicated system configurations and extensive evaluation. In this urban traffic environment the patterns of obstacle movements are very complicated and not always can be predicted. Another difficulty is that the host vehicle is moving, implying background's moving aspects. Strong limitations exist on the computational costs of the methods when considering the speed of the moving vehicle and the necessity of a real time system.

Some of the main characteristics which an obstacle detection system should have are: to be in real time, to be low price and compact in order to be easy incorporated in all vehicles on the road, to not interfere with other systems and the most important requirement is to be one hundred percent safety.

An important aspect of an obstacle detection and classification system from a traffic scene situation is its navigation. Thus a mobile autonomous system is mainly formed by a navigation process based on estimation and an update of the observations (performed at regular time intervals) of the mobile position related to the ruttier scene. Navigation systems are usually equipped with different sensors that can be used to track the position and describe the area of navigation.

2. Types of sensors

If we assume that a sensor may be defined as a device which measures some attribute of the world, then a classification of the sensors type can be according to the type of measurement information they give as output. Therefore, information provided from proprioceptive (measure an attribute regarding their own state) and exteroceptive (measure an attribute of an external object present in the scene) sensors can be mixed together in order to give a better perception of the environment.

2.1. Proprioceptive and exteroceptive sensors

Proprioceptive sensors sense the position, the orientation and the speed of the object to be detected. In the obstacle detection field, proprioceptive sensors monitor the motion of the vehicle by measuring the kinetic quantity like acceleration and velocity. In general, vehicles use inertial sensors like: accelerometers, tilt sensors, position sensors, odometers and speed sensors.

Exteroceptive sensors, like cameras, sonars, laser scanners and microwave radars, are employed to provide a representation of the environment (e.g. some visible features or the distance to the respective obstacles). These sensors give the vehicle information about the surrounding and that information allows the vehicle to interact with the world.

Distance sensors or proximity sensors perform the same task that vision and tactile sensing do in human beings. They are used to measure the relative distance (range) between the sensor and objects in the environment. For that, active sensors, infrared sensors or tactile sensors (bump sensors) can be used. Also, proximity measurements can be performed with the laser ranging, the usage of stereo cameras or projecting a pattern on the environment and observe how the pattern is distorted (the use of markers). In this paper we will assume that our obstacle detection system is traveling in an unknown environment, so the use of markers, maps (like those used with GPS (Global Positioning System)) or any other type of sensor which suppose the cognition of the surrounding environment are ignored.

Vision sensors work similarly to the eyes of the human beings: it refers the processing data using the electromagnetic spectrum (in general visible or infrared spectrum) to produce an image. In the obstacle detection and classification context it is used to recognize and classify objects by using CCD (Charge Coupled Device) cameras as vision sensors.

From technology's point of view, there are some different sensing technologies that can be used for obstacle detection. To record information about the obstacles presence on a road, active or passive sensors can be used. In the case of a passive sensor, the sensor just receives a signal. It detects the reflected, emitted or transmitted electro-magnetic radiation provided by natural energy sources. In the case of using an active sensor, the sensor emits a signal and receives a distorted copy of the respective signal. It detects reflected responses from objects irradiated with artificially generated energy sources.

2.2. Passive and active sensors

The sun's energy can be reflected (e.g. visible wavelengths), or absorbed and then reemitted (e.g. thermal infrared wavelengths). Passive detection can only work when the natural energy is available: for visible wavelength, as the sun provides the energy, sensors working in visible spectrum can be use only in daytime; thermal infrared sensors can work in day or night as long as the amount of energy is large enough to be recorded by the sensor's receiver.

Active sensors provide their own energy for illumination, and thus some possible interference problems can appear if they are deployed on many vehicles on the road. In general, in the obstacle detection domain, three different types of active sensors may be used: radar (RAdio Detection And Ranging), ladar (LAser Detection And Ranging) or lidar (Light-Imaging Detection And Ranging) and sonar (SOund Navigation And Ranging).

Passive sensors

From the passive sensors used in the obstacle detection field we will mention the visible spectrum camera (which capture the light from the visible electromagnetic spectrum) and the infrared camera (uses the infrared spectrum). These sensors most often are called vision-based because they register images. During the last several years, an impressive number of passive vision-based systems have been developed to perform the obstacle detection task, and they may be divided in two categories: one based on objects depth information (extracted using stereo images) and one based on objects motion information (extracted using video images at different time instances). Using a perspective translation in space respective in time, the resulted depth or motion information can be used to detect distance to the object.

Images captured by cameras working in visible spectrum are very rich in contents and easy to interpret. Maybe this is the reason why the most attention in research for obstacle detections was concentrated in this direction. Because visible spectrum sensors are passive (a natural energy source is required), they have limitations due to lighting conditions and possible shadings. The major issue is that they are not well suited for darkness conditions. Visible cameras do not emit any signals, and thus, an important advantage of this type of cameras, in addition to their low price, is that there are no interference problems with the environment.

The infrared spectrum is an excellent spectral region to use for object detection and tracking due to the fact that all heated objects emit infrared radiation that can be registered with an infrared camera. Since some classes of objects (pedestrians, vehicles) have a specific infrared signature (pedestrian's head, body, legs respective vehicle's wheel and engine), positive object identification can be made based on the received energy.

In the case of infrared spectrum sensors, we have to point out a specification: there are both optical infrared sensors active or passive. The active ones generates infrared wavelength light to detect reflections from possible target objects, while the passive ones just responds to changes in the received infrared light provided by the objects. The passive infrared sensors do not send and receive signals as do active sensors; instead, they passively wait until infrared energy from an object is detected. In the obstacle detection field, most often this type of infrared sensor is used in order to avoid any possible infrared radiation (no active illumination source is used). An infrared sensor system is a collection of optical elements and electronic hardware connected to an infrared detector. The optical elements reflect and focus incident radiation from an object onto a focal plane, and electronic hardware attached to the focal plane is used to read the electrical signals generated by each pixel of the focal plane. Signal processors are used to convert these analog voltage signals into digital images.

According to the infrared spectrum, three different types of infrared cameras exist: the near (NIR), middle (MIR) and far (FIR) infrared region. FIR cameras have been the most utilized infrared sensor in obstacle detection and in general the term "infrared" is referred for FIR, but during the last years, a lot of systems using also NIR spectrum appeared.

The infrared spectrum has some important advantages: have the ability to measure the temperature and because the passive ones are independent of the light source they can register the same or almost the same images even it is day or night. There are a lot of heats radiators apart from human, so infrared sensors can be used for obstacle detection in general not only for pedestrians. The main issues are that infrared sensors are sensitive to weather conditions (rain, fog) and infrared image processing is a relative new domain for obstacle detection.

Visible and infrared images differ especially because the visible spectrum camera registers reflected light while the infrared camera registers heat emitted in the scene. In an infrared image, bright regions correspond to heat, while in visible images bright regions correspond to amount of the reflected light.

Active sensors

Active sensors provide their own energy for illumination, and that energy is reflected by target objects in the scene. Therefore, the reflected energy provides information about the surrounding. The main advantages of using active sensors are their possibility to measure distance and speed of the target objects and the fact that they work well also in bad weather or poor illumination conditions. Besides the interference problems, other limitations of the active sensors are: difficulties in interpreting the output signal returned by these sensors and the acquisition price.

Radar is an active "radio detection and ranging" sensor which provides its own source of electromagnetic energy. An active radar sensor emits radiation in a series of pulses from an antenna. When the energy reaches the target, some of the energy is reflected back toward the sensor and this radiation is detected, measured, and timed. The time required for the energy to travel to the target and return back to the sensor determines the distance or range to the target. An advantage of radar is that images can be acquired day or night. Also, radar technology can operate in different environmental conditions (e.g. rain, snow, poor visibility) without any strong limitations.

The second active sensor, lidar ("light detection and ranging sensor") or ladar ("laser detection and ranging sensor") uses a laser to transmit a light pulse and a receiver to measure the reflected light. So, a lidar (ladar) is an active laser pulses for detecting reflections from objects. Distance to the object is determined by recording the time between transmitted and received pulses; the distance traveled is calculated using the speed of light. A laser scanner is a high-resolution laser range finder that emits infrared laser pulses and detects the reflected pulses. Having a high accuracy both in lateral and longitudinal direction, they provide precise measurement of depth and use large field of view. For processing lidar-images procedures similar to image processing are applied because the original data from a laser scanner is much like vision image data.

Sonar works very similar to radar: instead of electromagnetic radio or microwave, sound waves are transmitted from an antenna. Ultrasonic sensors generate ultrasonic waves which are reflected by targets. When an obstacle passes by, a portion of the transmitted sound wave is reflected back to the receiver. Based on its analysis, objects can be detected together with their distance and speed. One important disadvantage of sonar refers to weather conditions change, because the speed of sound waves varies according to temperature and pressure of the surrounding medium.

3. OBSTACLE DETECTION SYSTEMS

Vision systems

Video sensors provide a high spatial resolution, both horizontally and vertically. Almost all approaches using vision implies two steps: *object detection* followed by *object recognition.* In the detection step a region of interest (ROI) (e.g. a rectangular bounding box) is found and it is associated with a potential obstacle; then, the recognition process follows, where the type of objects is determined. Most often an object detection system is formed from three main components: one module which produces the segmentation of the objects from the background, another one which classify the objects and the last module which track the objects. Segmentation is a method to extract objects or features of the objects from the background. There are three segmentation methods used in the obstacle detection domain: stereo, motion or infrared based segmentation. The stereoscopy is a new developed technique used to extract three-dimensional information from bi-dimensional images. The extraction of depth information (also called stereo matching) is a process of finding correspondences between two correlated images [1], [2]. The main limit of the motion based segmentation technique is the need of processing a number of successive frames which increases the processing time [3]. The information extracted after segmentation can be a model or some features of the object. The system must be learned a priori with some features in order to detect and recognize these features. From the obtained images certain features are extracted (which will represent the pattern), and these patterns will be essentials for the learning and testing process.

An autonomous vehicle combines information provided from multiple sensors in order to obtain a better cognition of the surrounded. In this case, the provided vision data should come from: different sensors (visible and infrared), or from the same sensor (video or stereo images but taken at different moment of time or space) and they should be correlated each other. Different cameras or configurations are used in vision-based obstacle detection systems:

One possibility is that camera be in move (installed on a moving vehicle) or static (for surveillance purposes [4]). For video frames captured in intelligent vehicles the frame-background are always changing and unpredictable, and therefore no reference background (as in the case of static camera) can be used due to the motion of the objects inside the scene and the movement of the camera. Systems using moving cameras in visible, and in infrared domain are presented in [5,6,7,8] and [9,10,11].

The second possibility is the use of a single image or a sequence of images (video images). Single images are very rarely used alone for obstacle detection applications. The time processing is not very expensive, but either the performances of such a system are not very good. To recognize obstacles in single images can be a problem, because such a system is not using either the motion or stereo information. This is the case in which any obstacle from image is seen as a still object and for his detection shape-based approaches are used. Systems in which objects symmetry properties are exploited can be found in visible [5,12] and infrared [10,13] domain. Video images (or sequences of consecutive frames at increased moment of time) are processed in order to provide a motion vector, needed to identify moving objects.

A third possibility is the use of monocular or stereo images. Monocular images are often used for shape-based approaches in the visible domain [5,12,14] and in infrared domain [9,10,13]. This kind of images is also used in motion-based approaches in visible [15] and infrared [16] domain. One alternative solution to the monocular images is the use of stereo images, in visible [6,17] and infrared [11] domain.

Different algorithms are proposed to detect pedestrians in images acquired from far infrared (FIR) cameras [9,10,13,16]. Honda has developed an intelligent night vision system using two far infra-red cameras (one stereo system) which were installed in the front bumper of the vehicle.

Vision-based technique is still the easiest and utilized method to analyze the road, but more and more works appear in this domain and several of sensors-combination systems were developed. Almost all obstacle-detection systems use visible spectrum or infrared cameras in fusion with an active sensor, like radar, laser scanner or sonar to increase the performances of such a system.

Systems using active sensors

The use of radar for obstacle detection has advantages especially in bad weather situations. Radar sensor measures only point targets of objects like cars, trees, traffic signs, bicycles and human beings while other objects which does not have a high reflectance, like street and pavement, are not detected. In [18] and [19] some radar parameters like RCS (Radar Cross Section), power spectral density (PSD) and object's velocity coming from radar are registered (for pedestrians and vehicles) and compared in order to differentiate between these two types of objects.

The laser scanner measures the outline of objects with high resolution and accuracy. Object velocity is derived from the positional information of each scan [19]. The original data from a laser scanner is much like vision image. Clustered data points are grouped into different objects using segmentation and then the objects are classified into different categories (pedestrians, cars, trucks, bicycles) according to their characteristics. Some laser scanner characteristics are presented in [20], [21].

Sensor fusion

A much exploited idea is to combine some different sensors inputs together. In that way, one technology's advantage can be used to compensate another's limitation; the combination process is often called "sensor fusion". This sensorial fusion is done by humans every day, since humans combine visual with acoustic and tactile information to get more knowledge about the surrounding. Many times, the use of one single sensor is not very useful for an obstacle-detection system, because there is no such a perfect sensor which can provide rich information about the surrounding in any weather conditions and in any illumination situation. So it is a natural idea to make use of the advantages of one sensor to suppress another's disadvantages. A fusion process can take place on different levels: low or high level. Low level (data-level) means that the fusion is made at data information and combines results before a hard decision is made. High level fusion means that the fusion is established on a list of objects. It combines results of different modules after some decisions are made about the size and position of the obstacle. In general, in a high-level fusion, a list of objects is generated with one sensor and it is checked with help of the other sensor. Using different sensor technologies together, the system performances can be significantly improved. The sensor fusion increases the evaluation speed by reducing the search space (region of interest). Thus, one or more sensors of different technologies are bring together to perform a better recognition of the object. In general, because active sensors offer limited spatial information in opposite with a vision system which has rich information regarding visual appearance of an object, a fusion between passive and active sensors is desired.

Radar offer good accuracy in longitudinal distance measurement, but poor accuracy in lateral distance measurement and because vision has opposite properties, a fusion will bring good accuracy in both positional measurements. In [21] and [22] radar and vision sensors are used for a fusion application in dense city traffic situations. In [23] the sensor system for vehicles detection consists of a MMW (Milli-Meter Wave) radar and a camera sensor fusioned using a Bayesian network. Target car image is clipped out from whole image by using object position data acquired by MMW radar to reduce computational load of image processing. The main advantage of using IR camera instead of visible spectrum camera is that the segmentation problem is simplified. Sensors fusion using IR cameras and radar is suggested to support the driver in reduced visibility, due to night and adverse weather conditions in [24]. For the obstacle detection system presented in [25] data fusion is performed between stereovision and laser scanner. In [20] a multi sensor system that uses an infrared camera with a laser scanner is used. To handle the information from different sensors a Kalman filter based data fusion is used.

Where a single radar sensor installed on a vehicle may not be enough to cover the whole interested area, a radar network can be used. A high performance sensor platform for pedestrians and cyclists protection is presented in [26]. Their data come from: a radar network composed of several 12GHz radar sensors, an image system composed of passive infrared (IR) and an visible color vision systems is fused in a low and high level data fusion technique. A pedestrian detection system from a moving camera is proposed in [18]. The fusion of a radar system consisting of two individual 12 GHz radar sensors and a monocular vision system was performed. A target-list is generated from radar and the remained filtered pedestrian candidates are used as inputs for the signal-processing module of the vision sensor.

VI. CONCLUSION

A complete and reliable sensing system for obstacle detection can benefit a lot from the combined use of multiple types of sensors, especially from the active-passive combination. Any one particular type of technology may have difficulties to meet all necessary requirements in order to detect an obstacle in various lighting or weather conditions. The clutter background and complex moving patterns of all objects which may appear on a road scene demand sophisticated processing of sensor inputs.

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